All these examples illustrate one important aspect called "Steepness". The measure of steepness is called slope or gradient.

The concept of slope is important in economics because it is used to measure the rate at which the demand for a product changes in a given period of time on the basis of its price. Slope comprises of two factors namely steepness and direction.



Fig. 5.16

Definition

If θ is the angle of inclination of a non-vertical straight line, then $\tan\theta$ is called the slope or gradient of the line and is denoted by m.

Therefore the slope of the straight line is $m = \tan\theta$, $0 \leq \theta \leq 180°$, $\theta \neq 90°$

To find the slope of a straight line when two points are given

$$\text{Slope } m = \tan\theta$$

$$= \frac{\text{opposite side}}{\text{adjacent side}}$$

$$= \frac{BC}{AC}$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

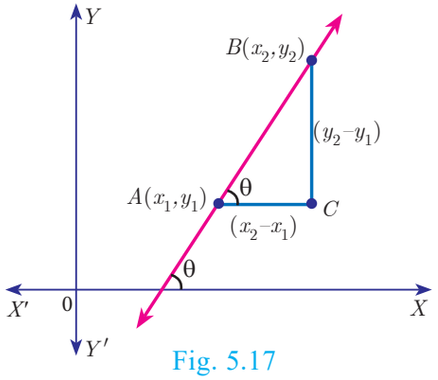


Fig. 5.17

$$\text{Slope } m = \frac{\text{Difference in } y \text{ coordinates}}{\text{Difference in } x \text{ coordinates}}$$

Note

The slope of a vertical line is undefined.

The slope of the line through (x_1, y_1) and (x_2, y_2) with $x_1 \neq x_2$ is $\frac{y_2 - y_1}{x_2 - x_1}$.

Values of slopes

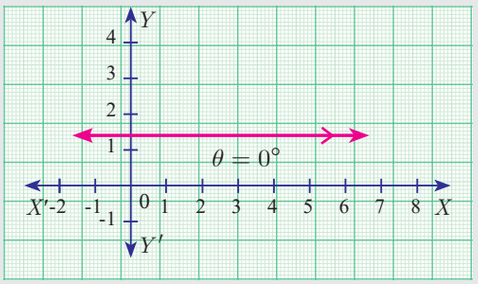
S. No.	Condition	Slope	Diagram
(i)	$\theta = 0°$	The line is parallel to the positive direction of X axis.	

Fig. 5.18(a)

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